Replication and Extension of a TDA-Based Sparse Index Tracking Model

Portfolio Optimisation Project

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Goel, A., Pasricha, P. and Kanniainen, J. (2025).
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Abstract

This study replicates the TDA-based sparse index tracking framework proposed by Goel et al. (2025) and extends it by introducing economically interpretable penalty measures. The original framework applies Topological Data Analysis (TDA) to asset return series and uses persistence landscape norms as asset-specific regularisation parameters within a weighted Elastic-Net optimisation problem. Although this approach delivers strong tracking performance, the economic meaning of the TDA penalties is not explicit, particularly in relation to systematic index exposure and asset-specific volatility.

Using S&P 500 data from Yahoo Finance, this study compares the original TDA models with a Beta–Volatility extension based on CAPM index beta and asset volatility. The models are evaluated through a rolling-window backtesting procedure across two sample periods. The results show that the TDA models are the strongest specifications for strict index replication, producing the lowest tracking errors and highest benchmark correlations. The BetaVol model, however, delivers higher annualised returns and lower turnover, but with higher volatility, VaR, and CVaR.

The findings confirm that TDA provides valuable structural information for sparse index tracking that is not fully captured by beta or volatility. At the same time, the Beta–Volatility framework offers a simpler and more transparent alternative when interpretability, lower turnover, and enhanced return exposure are prioritised over pure tracking accuracy.

Contents

1 Introduction

1.1 Project Definition

The passive investment strategy has witnessed remarkable growth over the past two decades, with global assets under management in exchange-traded funds (ETFs) surpassing US$11 trillion by 2024 (ETFGI 2024). The core assumption of passive investing is simple yet powerful: instead of attempting to outperform the market through active stock selection, investors seek to replicate the performance of a market benchmark such as the S&P 500. This problem is referred to as index tracking and involves constructing a portfolio whose returns closely mirror those of the target index.

In a recent breakthrough, Goel et al. (2025) introduced a novel methodology that leverages Topological Data Analysis (TDA) to determine asset-specific regularisation parameters for sparse index tracking. Their approach constructs persistence landscapes from asset return time series using Takens' embedding and employs the L_1 norms of zero-dimensional and one-dimensional persistence landscapes as the coefficients α_i and β_i within a weighted Elastic-Net formulation:

$$\sum_{t=1}^{T}(R_{w,t} - R_{0,t})^2 + \sum_{i=1}^{n} \alpha_i |w_i| + \sum_{i=1}^{n} \beta_i^2 w_i^2$$

Their empirical results, based on 23 years of S&P 500 data, demonstrate that TDA-based regularisation outperforms conventional methods such as Elastic-Net, LASSO, and SLOPE across multiple metrics, including tracking error, volatility, downside deviation, turnover, and risk-adjusted performance. Notably, the TDA-EN12 portfolio achieved a tracking error of 6.193×10^{-4} and a correlation of 99.3% with the index while maintaining lower turnover than traditional approaches.

1.2 The Unresolved Gap: What Does TDA Actually Measure?

Despite the strong empirical performance reported by Goel et al. (2025), the framework leaves a fundamental question unresolved: what economic risk does the TDA penalty actually represent? The persistence landscape norms aggregate all sources of return variability, including both systematic co-movement with the index and idiosyncratic asset-specific noise, into a single numerical quantity. Consequently, an asset with high beta and low idiosyncratic volatility may receive a similar penalty to an asset with low beta and high idiosyncratic volatility, provided that their topological complexities are comparable.

1.3 Background and Research Question of the Chosen Paper

This study builds on the work of Goel et al. (2025), who introduced a TDA-based methodology for sparse index tracking in their paper *Risk Reduced Sparse Index Tracking Portfolio: A Topological Data Analysis Approach*. Their study demonstrated that tools from persistent homology can be used to measure the structural riskiness of financial assets in a fully data-driven manner. These risk measures are then incorporated into a weighted Elastic-Net framework to guide portfolio construction.

The paper investigates whether a sparse portfolio can be constructed to closely replicate the performance of a market index while reducing portfolio risk and implementation costs. Traditionally, sparse index tracking has relied on regularisation techniques such as LASSO and Elastic-Net penalties. However, these approaches do not explicitly account for the structural behaviour of asset returns. Goel et al. (2025) therefore examine whether TDA can be used to construct more informative regularisation penalties that produce sparse index tracking portfolios which are lower risk, more cost-effective, and more accurate than conventional regularisation methods.

1.4 Research Question

While the approach of Goel et al. (2025) successfully captures the topological structure of asset returns, it does not explicitly incorporate two of the most fundamental measures of financial risk: an asset's sensitivity to the market and its own return volatility. This study therefore extends their framework by complementing the TDA-based persistence landscape norms with CAPM-derived index beta and asset volatility as regularisation parameters within the same weighted Elastic-Net framework.

Specifically, this study asks the following question:

> *Can using an asset's sensitivity to the index, measured through CAPM beta, together with its own volatility, as regularisation parameters within the weighted Elastic-Net framework of Goel et al. (2025), produce a sparse index tracking portfolio that is equally or more risk-efficient and tracks the index as well or better?*

In this study, risk-efficiency is not represented by a single optimisation objective or standalone numerical score, but rather is assessed through a joint evaluation of tracking performance, portfolio risk, and implementation efficiency. Specifically, portfolios are evaluated using tracking error and correlation with the benchmark index to measure replication accuracy, together with annualised volatility, maximum drawdown, Value-at-Risk (VaR), and Conditional Value-at-Risk (CVaR) to assess downside risk exposure.

In addition, Sharpe Ratios and Information Ratios are used to evaluate risk-adjusted performance, while turnover and the number of selected assets are included to account for implementation costs and portfolio sparsity. A portfolio is therefore considered more risk-efficient if it achieves comparable or lower tracking error while simultaneously maintaining lower risk, stronger risk-adjusted performance, reduced turnover, and a sparse but stable asset allocation.

1.5 Motivation

The paper by Goel et al. (2025) makes an important contribution to portfolio optimisation by addressing the sparse index tracking problem, which is discussed extensively in Cajas (2025, Chapter 9.2). This problem remains highly relevant because index replication is still constrained by transaction costs, overfitting risk, and instability of portfolio weights in high-dimensional financial datasets (DeMiguel et al. 2024). As a result, sparse index tracking has become a major focus in modern portfolio optimisation research.

Goel et al. (2025) extend traditional index tracking methods by incorporating TDA-based risk measures into a weighted Elastic-Net framework, allowing portfolio sparsity to be guided by the structural properties of asset returns rather than purely statistical penalties. This shift is motivated by evidence that conventional LASSO and Elastic-Net approaches may become unstable in the presence of highly correlated financial assets, resulting in poor out-of-sample performance (Fan & Liao 2022).

However, while TDA captures complex structural information, it is less directly interpretable in terms of standard asset pricing concepts such as systematic and idiosyncratic risk (Su et al. 2025). To address this limitation, this study proposes an extension based on the Capital Asset Pricing Model (CAPM), where the market portfolio is replaced with the target index so that beta directly measures each asset's exposure to index movements. This connects naturally with the risk-factor framework discussed in Cajas (2025, Chapter 4).

Volatility is additionally incorporated to capture residual risk not explained by beta, linking directly to the convex risk measures discussed in Cajas (2025, Chapter 7). CAPM-based frameworks continue to be widely used to measure systematic risk through beta and remain relevant in modern optimisation settings (Brusov et al. 2024, Flández et al. 2025). Similarly, volatility remains an important complementary risk measure in empirical portfolio construction because it captures idiosyncratic risk not explained by systematic exposure (Kim et al. 2023).

These developments motivate a hybrid framework that combines CAPM-based beta, using the index as the market proxy, together with volatility inside an Elastic-Net structure to improve both interpretability and risk control in sparse index tracking portfolios.

1.6 Contributions of This Study

This study contributes to the work of Goel et al. (2025) in several ways.

First, it provides one of the first empirical tests of whether economically interpretable CAPM-based penalties can match or complement the performance of TDA-based regularisation methods. If successful, the proposed approach offers practitioners a computationally simpler and more transparent alternative to topological methods.

Second, the study introduces an asymmetric penalty structure that treats high-beta assets as desirable for index tracking, representing a conceptual departure from conventional regularisation methods that penalise all forms of variability equally. This aligns the optimisation objective more closely with the economic intuition of benchmark replication.

Third, the study evaluates portfolios not only in terms of tracking accuracy, but also through a composite risk-efficiency framework incorporating volatility and drawdown protection. This addresses practical concerns faced by institutional investors that are not explicitly considered in Goel et al. (2025).

The remainder of this paper is organised as follows. Section 2 reviews the relevant literature on sparse index tracking, regularisation methods, Topological Data Analysis, and CAPM-based risk measures. Section 3 presents the methodology used to replicate the framework of Goel et al. (2025), including the rolling-window procedure, optimisation framework, and evaluation metrics. Section 4 introduces the proposed Beta–Volatility extension and describes the construction of the CAPM-based penalty structure. Section 5 presents the empirical results and compares the proposed framework against the TDA benchmark models across both calm and turbulent market periods. Section 6 discusses the main findings, interpretations, and implications of the results. Finally, Section 7 concludes the study, outlines its limitations, and suggests directions for future research.

2 Literature Review

2.1 Index Tracking and Sparse Portfolio Construction

Index tracking, also known as passive investing or benchmark replication, has become one of the most significant areas in modern quantitative finance. The objective is to construct a portfolio whose returns closely follow those of a benchmark index, such as the S&P 500, while minimising transaction costs, operational complexity, and portfolio risk. This approach supports the rapid growth of exchange-traded funds (ETFs) and index-linked investment products, and has attracted sustained attention across financial engineering, statistics, and data science (Malhotra 2024). Empirical evidence shows that the difficulty of consistently generating excess returns, or alpha, through active management has further reinforced the shift toward passive strategies, particularly in large and liquid markets (Prondzinski & Miller 2024). As a result, index tracking has evolved from a practical implementation problem into a cornerstone topic in both academic research and practical portfolio management.

In its simplest form, index tracking can be implemented through full replication, where each constituent asset is held in proportion to its benchmark weight. As indicated by Dhingra et al. (2026), while this minimises tracking error, it is rarely feasible in practice for large indices such as the S&P 500, where the presence of illiquid securities, transaction costs, and operational constraints makes full replication excessively expensive. These limitations have led to the development of sparse index tracking, where the portfolio is constructed using only a subset of available assets while preserving the key return dynamics of the benchmark. This introduces a fundamental trade-off between tracking accuracy and implementation cost, a theme that has been widely explored in the literature (Che et al. 2022). This trade-off is directly addressed in Cajas (2025, Chapter 9.2), which covers index tracking as a constrained portfolio optimisation problem and motivates the need for sparsity-inducing methods that balance tracking precision with practical implementability.

Early contributions by Roll (1992) and Pope & Yadav (1994) established the relationship between tracking error, active risk, and portfolio efficiency, laying the foundation for subsequent work on constrained portfolio optimisation. Alongside these, rule-based approaches based on market capitalisation filters, correlation screening, and clustering were proposed to reduce dimensionality before optimisation. Zhang & De Smedt (2024) and Yi et al. (2022) demonstrate that clustering financial time series can reveal natural groupings of assets, facilitating efficient index replication. However, these approaches lack formal optimality guarantees and do not explicitly control tracking error. Cardinality-constrained formulations offer a more direct alternative but are shown to be NP-hard as the number of assets increases (Jo & Cho 2025), motivating the shift toward regularisation-based approaches discussed in the following section.

2.2 Regularisation-Based Approaches to Sparse Index Tracking

To overcome the computational challenges of cardinality constraints, the literature has shifted toward regularisation-based approaches that induce sparsity through penalisation. LASSO, adapted to portfolio optimisation by Brodie et al. (2009), penalises the absolute magnitude of portfolio weights to simultaneously reduce tracking error and encourage sparse solutions. The Elastic-Net combines L_1 and L_2 penalties to address the issue of highly correlated assets common in financial datasets, improving out-of-sample performance (Yao & Zhang 2023). These methods are directly covered in Cajas (2025, Chapter 9.2). However, as indicated by Kremer et al. (2020), under portfolio constraints where weights must sum to one and remain non-negative, the L_1 norm becomes constant and loses its sparsity-inducing effect. This motivated the development of adaptive regularisation techniques with asset-specific penalty weights, including the adaptive Elastic-Net (Shu et al. 2020) and Sorted L-One Penalised Estimator (SLOPE), which assigns decreasing penalties to ordered coefficients and has demonstrated improved sparsity and reduced turnover (Kremer et al. 2022). More recently, group LASSO methods use sector information to improve sparse portfolio selection by identifying important sectors and assets simultaneously (Sadik et al. 2024), while Xia et al. (2023) demonstrate that high-dimensional sparse portfolio selection under non-negativity constraints remains an active and evolving area of research.

A persistent challenge across all these methods is the selection of regularisation parameters, with cross-validation remaining unreliable in financial time series due to autocorrelation and structural breaks (Cerqueira et al. 2020). It is this gap, the need for data-driven and economically grounded parameter calibration, that Goel et al. (2025) address through TDA, and that this study seeks to complement through CAPM-based beta and volatility.

2.3 Risk Measurement and Risk Efficiency

While traditional index tracking formulations focus on minimising tracking error, this alone does not adequately capture downside risk, particularly during periods of market stress (Biswas & Sen 2025). This study defines risk-efficiency as a composite concept encompassing portfolio volatility, Value-at-Risk (VaR), Conditional Value-at-Risk (CVaR), and downside deviation, while simultaneously preserving tracking accuracy and limiting turnover as a proxy for transaction costs. This definition is motivated by evidence that tracking error minimisation alone is insufficient for capturing the full risk profile of a portfolio during market stress (Biswas & Sen 2025), and that downside risk measures such as CVaR provide theoretically superior risk characterisation as coherent risk measures, as discussed in Cajas (2025, Chapter 7) and Rockafellar & Uryasev (2000). Recent evidence confirms that incorporating downside risk measures into portfolio construction delivers improved risk-adjusted performance, particularly during turbulent market conditions (Che et al. 2022, Jo & Cho 2025).

2.4 Topological Data Analysis

The limitations of traditional risk measures have motivated the exploration of non-parametric approaches. Topological Data Analysis (TDA) provides a framework for capturing the structural properties of financial time series through persistent homology, identifying features that persist across multiple scales. Recent work by Akingbade et al. (2024) and Rudkin et al. (2023) demonstrates that topological indicators can detect early warning signals of market instability and structural changes not reflected in traditional volatility measures, with Akingbade (2026) further confirming that TDA captures structural patterns in financial time series that conventional methods overlook. A change point detection study applying TDA to 26 stocks over 12 years finds that persistence landscape norms serve as reliable financial volatility indicators (Yao et al. 2025).

Goel et al. (2025) build directly on this literature by applying Takens' embedding and Vietoris–Rips filtration to extract topological features from asset return series. They compute persistence landscapes whose L_1 norms serve as asset-specific regularisation parameters in a weighted Elastic-Net framework, providing a fully data-driven, model-free approach to parameter calibration without expensive cross-validation.

2.5 CAPM-Based Risk Factors and Volatility

While the TDA framework of Goel et al. (2025) is methodologically novel, it captures risk purely from a topological perspective without reference to standard asset pricing theory, making it less directly interpretable in terms of systematic and idiosyncratic risk (Akingbade 2026). This study does not seek to replace TDA but rather to complement it by proposing an alternative regularisation framework grounded in classical financial theory. The Capital Asset Pricing Model (CAPM), covered in Cajas (2025, Chapter 4), establishes that an asset's expected return is a function of its sensitivity to a systematic risk factor.

In the index tracking context, the conventional market portfolio is replaced with the target index so that the estimated beta directly measures each asset's co-movement with the benchmark. Assets whose beta deviates from one introduce tracking error and are penalised more heavily, while those with beta close to one are favoured as natural candidates for inclusion. CAPM-based frameworks continue to be widely used in modern optimisation settings (Flández et al. 2025). Asset volatility is incorporated as a complementary measure of idiosyncratic risk not explained by systematic exposure, consistent with Cajas (2025, Chapter 4), and remains a key risk signal in empirical portfolio construction (Rujivan et al. 2025). Together, index beta and volatility provide a two-dimensional, economically interpretable risk characterisation that jointly replaces the TDA-based persistence norms as regularisation parameters in the weighted Elastic-Net framework of Goel et al. (2025).

3 Methodology

This section explains the methodology used in the study. The work is divided into two main parts. The first part replicates the TDA-based sparse index tracking approach of Goel et al. (2025). The second part introduces an economic extension, where CAPM beta and volatility are used as alternative penalty coefficients in the same weighted Elastic-Net framework.

The purpose of the extension is not to claim that beta and volatility are more advanced than TDA. Instead, the aim is to test whether simpler and more interpretable financial risk measures can provide useful sparse index tracking performance when placed in the same optimisation structure. This allows us to compare topological risk measures against standard financial risk measures in a fair setting.

3.1 Data and Experimental Setup

Since the original data used by Goel et al. (2025) was sourced from Thomson Reuters Datastream, which is not accessible for this study, the empirical analysis is conducted using an alternative but comparable dataset. Specifically, daily closing prices of the S&P 500 index and its constituent stocks, such as, amongst others, AAPL, MSFT, AMZN, GOOGL, and META, from Yahoo Finance were collected, a widely used and freely accessible source of financial market data. Following the work of Goel et al. (2025), the dataset is divided into two distinct panels:

- **Panel I:** 11 October 1999 to 10 October 2018 (long-term period)
- **Panel II:** 1 January 2018 to 8 March 2023 (COVID-19 and recent period)

Only constituents with complete price data throughout each sub-period are retained, with stocks containing missing observations dropped from the analysis. This procedure yields 355 constituents in Panel I and 477 constituents in Panel II, compared to 392 and 474 reported by Goel et al. (2025). The difference is attributable to the use of Yahoo Finance rather than Thomson Reuters Datastream as the data source. This discrepancy warrants careful consideration. To ensure full transparency and replicability, the exact list of retained and dropped tickers for each sub-period is documented in the GitHub. Furthermore, we acknowledge that stocks with missing data are more likely to include smaller, more volatile, or less liquid constituents, meaning our surviving universe may be systematically biased toward larger, more stable stocks Ranse (2026)

Daily arithmetic returns are computed as

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

where $P_{i,t}$ is the adjusted closing price of asset i on day t.

A rolling-window framework is used throughout the study. Each in-sample window contains 504 trading days, which is approximately two years. The portfolio is then evaluated over the next 21 trading days, which is approximately one month. The window is shifted forward by 21 days each time. This gives 193 rolling windows for Period I and 38 rolling windows for Period II.

At each rolling window, all model parameters are estimated using only the in-sample data. The resulting portfolio weights are then applied to the next out-of-sample period. This creates a realistic walk-forward testing setup.

3.2 TDA-Based Sparse Index Tracking

3.2.1 Takens Embedding

The TDA model starts by converting each one-dimensional return series into a point cloud. This is done using Takens' delay embedding. For a return series $\{x_t\}_{t=1}^{T}$, the embedded point cloud is given by

$$\mathbf{X}_i = \left\{ (x_t, x_{t+\tau}, \ldots, x_{t+(d-1)\tau}) \right\}_{t=1}^{T-(d-1)\tau}. \tag{1}$$

Following the original paper, we use embedding dimension $d = 3$ and delay $\tau = 1$. Each 504-day in-sample window is divided into overlapping sub-series of length $\tilde{T} = 42$ days with a step size of $h = 21$ days. Therefore, each asset produces 23 overlapping sub-series per rolling window. Each sub-series is embedded separately. This allows the TDA method to capture local structure in the return dynamics rather than using only one point cloud for the full two-year window.

3.2.2 Persistent Homology

For each embedded point cloud, a Vietoris–Rips complex is constructed over increasing scale values. Persistent homology is then used to track topological features as the scale changes. Two homology dimensions are considered:

- H_0, which captures connected components and clustering structure;
- H_1, which captures loop-like or cyclic structure.

The output is a persistence diagram, where each point records the birth and death of a topological feature. Infinite death values and invalid birth-death pairs are removed. If a diagram is empty after filtering, the placeholder point $(0, 0)$ is used for numerical stability.

3.2.3 Persistence Landscape Norms

Persistence diagrams are converted into persistence landscapes using GUDHI. The first landscape layer is used, which is consistent with the implementation described in the original TDA framework. The landscape resolution is set to 100. For each asset, the landscapes are averaged across the 23 sub-series.

The L^1 norm of the mean H_0 landscape is used as the first TDA penalty coefficient:

$$\alpha_i^{\text{TDA}} = \left\| \lambda_i^{(0)} \right\|_1 = \int |\bar{\lambda}_{0,i}(t)| \, dt. \tag{2}$$

The L^1 norm of the mean H_1 landscape is used as the second TDA penalty coefficient:

$$\beta_i^{\text{TDA}} = \left\| \lambda_i^{(1)} \right\|_1 = \int |\bar{\lambda}_{1,i}(t)| \, dt. \tag{3}$$

Numerically, these integrals are approximated using the trapezoidal rule with step size

$$\Delta x = \frac{1}{\text{resolution}}.$$

The intuition is that larger persistence landscape norms indicate more complex or less stable return behaviour. In the optimisation, such assets receive stronger penalties.

3.3 Weighted Elastic-Net Optimisation

Let $\mathbf{R}_{\text{in}} \in \mathbb{R}^{T \times n}$ be the in-sample matrix of stock returns and let $\mathbf{r}_{0,\text{in}} \in \mathbb{R}^T$ be the in-sample index return vector. The portfolio weight vector is denoted by $\mathbf{w} \in \mathbb{R}^n$.

The general weighted Elastic-Net index tracking problem is

$$\min_{\mathbf{w}} \|\mathbf{R}_{\text{in}}\mathbf{w} - \mathbf{r}_{0,\text{in}}\|_2^2 + \sum_{i=1}^{n} a_i |w_i| + \sum_{i=1}^{n} b_i^2 w_i^2, \qquad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1. \tag{4}$$

The first term is the tracking error term. It makes the portfolio follow the benchmark index. The second term is the weighted LASSO penalty. It encourages sparsity by pushing some asset weights to zero. The third term is the weighted ridge penalty. It shrinks large weights and stabilises the portfolio.

In this study, the same optimisation structure is used for both the TDA models and the economic extension. The only difference is how the penalty coefficients a_i and b_i are constructed.

3.4 TDA Model Specifications

The TDA replication uses four model specifications. These models differ only in how the penalty coefficients are assigned.

Model	a_i	b_i	**Description**
TE	0	0	No penalty; pure tracking error benchmark
TDA-Lasso	α_i^{TDA}	0	TDA H_0 norm used only in the LASSO term
TDA-EN11	α_i^{TDA}	α_i^{TDA}	TDA H_0 norm used in both LASSO and ridge terms
TDA-EN12	α_i^{TDA}	β_i^{TDA}	TDA H_0 used in LASSO and TDA H_1 used in ridge

The TE model is the unregularised benchmark. TDA-Lasso tests whether the H_0 landscape norm alone can produce a sparse tracking portfolio. TDA-EN11 uses the same H_0 norm for both selection and shrinkage. TDA-EN12 is the richer TDA model because it uses H_0 for sparsity and H_1 for shrinkage.

3.5 Performance Evaluation

For each rolling window, the estimated weights are applied to the next 21-day out-of-sample return matrix:

$$r_{p,t}^{(k)} = r_{oos,t}^{(k)\top} w^{(k)}. \tag{5}$$

The out-of-sample returns from all windows are then concatenated to form the final portfolio return series. The models are evaluated using tracking, risk, sparsity, and stability measures.

3.5.1 Tracking and Return Measures

Tracking error is calculated as

$$TE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(r_{p,t} - r_{0,t})^2}. \tag{6}$$

Correlation is computed between portfolio returns and index returns. A high correlation indicates that the portfolio moves closely with the benchmark.

Annualised return is computed as

$$\mu_p^{ann} = \bar{r}_p \times 252, \tag{7}$$

and annualised volatility is computed as

$$\sigma_p^{ann} = std(r_p)\sqrt{252}. \tag{8}$$

The Sharpe ratio is computed as

$$SHR = \frac{\mu_p^{ann}}{\sigma_p^{ann}}, \tag{9}$$

where the risk-free rate is taken as zero for simplicity.

The information ratio is computed as

$$IR = \frac{\overline{(r_p - r_0)} \times 252}{std(r_p - r_0)\sqrt{252}}. \tag{10}$$

3.5.2 Risk Measures

Maximum drawdown is the largest peak-to-trough fall in the cumulative wealth process.

Downside deviation is measured relative to the index:

$$DD = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \max(r_{0,t} - r_{p,t}, 0)^2}. \tag{11}$$

The empirical 95% Value-at-Risk is computed from the 5% quantile of out-of-sample portfolio returns:

$$VaR_{0.95} = -Q_{0.05}(r_p). \tag{12}$$

The Conditional Value-at-Risk is the average loss beyond the VaR threshold:

$$CVaR_{0.95} = -\mathbb{E}[r_p \mid r_p \leq Q_{0.05}(r_p)]. \tag{13}$$

3.5.3 Sparsity, Turnover, and Gross Exposure

The average number of selected assets is calculated as the average number of weights larger than 10^{-8} in absolute value:

$$\#Assets = \frac{1}{K}\sum_{k=1}^{K}\sum_{i=1}^{n}\mathbf{1}\{|w_i^{(k)}| > 10^{-8}\}. \tag{14}$$

Turnover is calculated as

$$TR = \frac{1}{K-1}\sum_{k=1}^{K-1}\left\|w^{(k+1)} - w^{(k)}\right\|_1. \tag{15}$$

Gross exposure is calculated as

$$GE^{(k)} = \sum_{i=1}^{n}|w_i^{(k)}|. \tag{16}$$

Gross exposure is important because the base optimisation only imposes $1^\top w = 1$. If gross exposure is much larger than one, the model may be relying on large long-short positions. If it is close to one, the portfolio is effectively non-leveraged.

3.6 VaR Backtesting: Kupiec Test

To formally check VaR performance, the Kupiec unconditional coverage test is used. For each rolling window, a historical 5% VaR threshold is estimated from the in-sample portfolio returns. This threshold is then applied to the next out-of-sample window.

A VaR violation occurs when

$$r_{p,t} < \hat{v}_{0.05}^{(k)}. \tag{17}$$

Let T_1 be the number of violations and let $T_0 = T - T_1$ be the number of non-violations. The empirical failure rate is

$$\hat{p} = \frac{T_1}{T}. \tag{18}$$

The Kupiec likelihood ratio statistic is

$$LR_{UC} = -2\left[T_0\ln(1-\alpha) + T_1\ln(\alpha) - T_0\ln(1-\hat{p}) - T_1\ln(\hat{p})\right], \tag{19}$$

where $\alpha = 0.05$. Under the null hypothesis of correct unconditional VaR coverage,

$$LR_{UC} \sim \chi_1^2. \tag{20}$$

A small p-value means that the number of VaR failures is not consistent with the expected 5% failure rate.

4 Extension: A Novel Contribution

The economic extension compares the TDA-based persistence landscape norms against CAPM-derived beta and volatility penalties within the same weighted Elastic-Net framework. The purpose is to test whether a simpler and more interpretable penalty design can compete with the TDA-based approach

4.1 Parameter Construction

For each asset i within each rolling window, two risk measures are estimated from the same in-sample return data already used in Goel et al. (2025).

4.1.1 Index Beta (Systematic Risk)

Beta is estimated by replacing the conventional market portfolio in the CAPM with the target index, so that beta directly measures each asset's sensitivity to the benchmark being tracked:

$$\hat{\beta}_i^{\text{CAPM}} = \frac{\text{Cov}(R_i, R_0)}{\text{Var}(R_0)}$$

where R_i is the return of asset i and R_0 is the return of the S&P 500 index. This modification is theoretically motivated in the index tracking context (Cajas 2025). What matters is not an asset's sensitivity to an abstract market portfolio, but rather its co-movement with the specific benchmark being replicated.

4.1.2 Asset Volatility (Idiosyncratic Risk)

Asset volatility is estimated as the annualised standard deviation of daily returns from the same rolling window:

$$\sigma_i = \sqrt{\text{Var}(R_i)}$$

This captures the overall variability of each asset's returns, serving as a measure of idiosyncratic risk not explained by systematic index exposure.

4.2 Penalty Specification

The regularisation parameters are defined as:

$$\alpha_i = \frac{1}{\left|\hat{\beta}_i^{CAPM}\right| + \varepsilon}, \qquad b_i = \sigma_i$$

where ε is a small positive constant included solely to ensure numerical stability when beta is close to zero, preventing division by zero.

The use of the reciprocal transform

$$\frac{1}{|\beta_i| + \varepsilon}$$

rather than alternative transforms such as

$$1 - |\beta_i|$$

is motivated by the following considerations.

First, the reciprocal function produces a strongly convex and asymmetric penalisation scheme. As beta approaches zero, the penalty grows without bound, forcefully excluding assets that have virtually no co-movement with the index. This is economically desirable in index tracking, where an asset with

$$\beta_i \approx 0$$

contributes almost nothing to replicating the benchmark and should be strongly discouraged from entering the portfolio.

Second, the reciprocal transform produces larger penalties for low-beta assets in a smooth and differentiable manner, which is compatible with the convex optimisation framework of the Elastic-Net.

Third, while

$$1 - |\beta_i|$$

would produce a similar directional effect, it assigns a penalty of zero to assets with

$$|\beta_i| = 1$$

which is precisely the most desirable asset for index tracking, whereas the reciprocal assigns a penalty of approximately 1 to such assets, maintaining a non-zero but minimal penalty that preserves the regularisation structure of the Elastic-Net.

Setting

$$b_i = \sigma_i$$

and using b_i^2 in the quadratic ridge penalty penalises high-volatility assets, reducing their weight in the portfolio and thereby lowering overall portfolio volatility.

4.3 Scaling of Economic Penalties

The beta-based and volatility-based penalties have different numerical scales. For this reason, both are scaled by their cross-sectional means within each rolling window:

$$\alpha_i^{\text{econ}} = \frac{\tilde{\alpha}_i^{\text{econ}}}{\frac{1}{n}\sum_{j=1}^{n} \tilde{\alpha}_j^{\text{econ}}}, \tag{21}$$

$$b_i^{\text{econ}} = \frac{\hat{\sigma}_i}{\frac{1}{n}\sum_{j=1}^{n} \hat{\sigma}_j}. \tag{22}$$

This scaling makes the average penalty equal to one in each window. It does not make all penalties equal. The individual penalty values still differ across stocks, but the average scale is controlled. This prevents one penalty type from dominating only because of its measurement units.

4.4 Economic Model Specifications

Four economic models are estimated.

Model	a_i	b_i	**Description**
Vol-EN11	b_i^{econ}	b_i^{econ}	Volatility used in both LASSO and ridge terms
Beta-EN11	α_i^{econ}	α_i^{econ}	Inverse-beta used in both LASSO and ridge terms
BetaVol	α_i^{econ}	b_i^{econ}	Inverse-beta used in LASSO; volatility used in ridge
BetaVol-Long	α_i^{econ}	b_i^{econ}	BetaVol with the extra constraint $w_i \geq 0$

The main extension is BetaVol. In this model, beta guides asset selection through the LASSO term, while volatility controls weight stability through the ridge term. Vol-EN11 and Beta-EN11 are included as ablation or sensetivity analysis models, since they help show whether beta or volatility is driving the results. BetaVol-Long is included as a robustness model to check whether the results remain valid under a long-only constraint.

The BetaVol optimisation problem is

$$\min_{\mathbf{w}} \|\mathbf{R}_{\text{in}}\mathbf{w} - \mathbf{r}_{0,\text{in}}\|_2^2 + \sum_{i=1}^{n} \alpha_i^{\text{econ}}|w_i| + \sum_{i=1}^{n} \left(b_i^{\text{econ}}\right)^2 w_i^2, \qquad \text{s.t.} \quad \mathbf{1}^\top\mathbf{w} = 1. \tag{23}$$

For BetaVol-Long, the additional constraint

$$w_i \geq 0, \qquad i = 1, \ldots, n \tag{24}$$

is imposed.

No additional Parameters. This is done to remain close to the data-driven penalty structure of the TDA framework. The penalty coefficients are constructed directly from the data in each rolling window and then used inside the same weighted Elastic-Net objective.

4.5 Optimisation Problem

The portfolio weights are obtained by solving the following optimisation problem:

$$\min_{w} \sum_{t=1}^{T}(R_{w,t} - R_{0,t})^2 + \sum_{i=1}^{n} \frac{1}{|\hat{\beta}_i| + \varepsilon}|w_i| + \sum_{i=1}^{n} \sigma_i^2 w_i^2$$

The asset-specific parameters

$$\alpha_i = \frac{1}{|\hat{\beta}_i| + \varepsilon}$$

and

$$b_i = \sigma_i$$

are pre-computed from the in-sample data before optimisation is performed, making the objective function a standard weighted Elastic-Net problem that can be solved efficiently using convex optimisation solvers.

4.6 Sensitivity Analysis for ε

The inverse-beta penalty depends on the small constant ε. This constant prevents the penalty from becoming infinite when beta is close to zero. However, the choice of ε can affect the penalty assigned to low-beta stocks.

To check that the results are not driven by one arbitrary value of ε, the BetaVol model is tested using

$$\varepsilon \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}. \tag{25}$$

The full rolling backtest is repeated for each value. The resulting tracking error, correlation, Sharpe ratio, VaR failure rate, number of selected assets, and gross exposure are compared. If the results remain similar across these values, then the model is not highly sensitive to the stabilisation constant.

4.7 Rolling Window Implementation

The same rolling window framework as Goel et al. (2025) is employed:

1. For each rolling window, estimate $\hat{\beta}_i$ and σ_i for all assets i using in-sample data.
2. Pre-compute the asset-specific penalty parameters

$$\alpha_i = \frac{1}{|\hat{\beta}_i| + \varepsilon} \quad \text{and} \quad b_i = \sigma_i$$

3. Solve the optimisation problem to obtain portfolio weights w.
4. Apply the estimated weights to the subsequent 21-day out-of-sample period.
5. Record realised portfolio returns.
6. Shift the window forward by 21 days and repeat the process.

4.8 Evaluation Metrics

The performance of the Beta-Volatility Elastic-Net model is evaluated against both the original TDA-based models and all benchmark methods of Goel et al. (2025) using the following metrics:

4.8.1 Tracking Performance

- **Tracking Error (TE)** - root mean squared deviation from index returns
- **Correlation** - Pearson correlation between portfolio and index returns

4.8.2 Risk Measures

- **Annualized Volatility** - standard deviation of portfolio returns
- **Maximum Drawdown** - largest peak-to-trough loss over the out-of-sample period
- **Value-at-Risk (VaR)** - maximum potential loss at 95% confidence level

- **Conditional VaR (CVaR)** - expected loss beyond VaR, also known as Expected Shortfall

4.8.3 VaR Backtesting

In addition to reporting VaR and CVaR estimates, we implement a Kupiec (1995) proportion of failures test to formally backtest the realised VaR violations of each portfolio. The Kupiec test assesses whether the proportion of days on which portfolio losses exceed the VaR estimate is statistically consistent with the chosen confidence level, providing a rigorous and formal validation of the VaR estimates that goes beyond simple point estimation. This is directly consistent with the VaR backtesting framework discussed in Cajas (2025) Chapter 15 and addresses the formal statistical validation requirement of the evaluation.

4.8.4 Risk-Adjusted Performance

- **Sharpe Ratio** - excess return per unit of volatility
- **Information Ratio** - excess return relative to the index per unit of tracking error

4.8.5 Transaction Costs

- **Turnover Ratio** - average absolute change in portfolio weights across rebalancing windows
- **Number of Assets Selected** - measures portfolio sparsity

All metrics are computed using the concatenated out-of-sample return series obtained from the rolling window procedure, consistent with the evaluation approach of Goel et al. (2025).

4.9 Penalty Comparison Between TDA and Economic Measures

To understand whether TDA and the economic penalties measure similar or different forms of risk, a penalty comparison is performed. The comparison is done across all rolling windows, not only one selected window.

Two comparisons are made:

- TDA H_0 penalty α_i^{TDA} versus economic inverse-beta penalty α_i^{econ};

- TDA H_1 penalty β_i^{TDA} versus economic volatility penalty b_i^{econ}.

Both Pearson and Spearman correlations are computed. Pearson correlation measures linear association, while Spearman correlation measures whether the rankings of assets are similar. The correlations are computed using pooled window-asset observations and also window-by-window.

If the correlations are high, this suggests that TDA and economic penalties capture similar information. If the correlations are low, this suggests that the TDA measures contain information not captured by beta and volatility.

4.10 Limitations of the Economic Extension

The economic extension is easier to interpret and much cheaper to compute than the TDA pipeline. However, it also loses some information. CAPM beta is a linear measure of co-movement with the index, while volatility is a second-moment risk measure. These two quantities do not capture nonlinear return geometry, regime shifts, clustering, or cyclical structures that may be detected by persistent homology.

Therefore, the economic extension should not be viewed as a direct replacement for the TDA model. Instead, it is treated as an interpretable competing benchmark. If the economic model performs close to the TDA model, this suggests that simpler financial measures may be sufficient for some sparse tracking tasks. If the TDA model performs better, this suggests that the topological information adds value beyond beta and volatility.

5 Empirical Results

This section presents the empirical results for the TDA-based models and the proposed economic Beta–Volatility extensions. The models are evaluated over two periods. Period I covers the long-run sample from 2000 to 2018, while Period II covers the more recent period from 2018 to 2023. All results are based on the out-of-sample returns generated from the rolling-window backtesting procedure.

5.1 Out-of-Sample Performance

Tables 1–4 report the out-of-sample performance of the TDA-based models and the proposed economic extensions. The results are split into smaller tables to improve readability.

Table 1: Out-of-sample performance: Period I — Tracking and return measures

Model	**TError**	**Correlation**	**Ann.Return**	**Volatility**
TE	0.001678	0.990470	0.122036	0.190308
TDA-Lasso	0.001627	0.990237	0.126537	0.183480
TDA-EN11	0.001626	0.990241	0.126535	0.183480
TDA-EN12	0.001627	0.990237	0.126537	0.183480
Vol-EN11	0.007474	0.772021	0.137302	0.123399
Beta-EN11	0.010861	0.922685	0.191472	0.326994
BetaVol	0.008579	0.946875	0.186003	0.296954
BetaVol-Long	0.008579	0.946875	0.186003	0.296954

Table 2: Out-of-sample performance: Period I — Risk and downside measures

Model	**MaxDD**	**DownsideDev**	**VaR**	**CVaR**
TE	-0.518142	0.001053	0.016864	0.028679
TDA-Lasso	-0.521554	0.000991	0.016650	0.027583
TDA-EN11	-0.521544	0.000991	0.016650	0.027583
TDA-EN12	-0.521554	0.000991	0.016650	0.027583
Vol-EN11	-0.299109	0.005146	0.010982	0.017616
Beta-EN11	-0.647774	0.007373	0.031197	0.048358
BetaVol	-0.623272	0.005720	0.027882	0.044375
BetaVol-Long	-0.623271	0.005720	0.027882	0.044375

Table 3: Out-of-sample performance: Period II — Tracking and return measures

Model	TError	Correlation	Ann.Return	Volatility
TE	0.003606	0.975518	0.103410	0.260077
TDA-Lasso	0.002019	0.991917	0.135600	0.251964
TDA-EN11	0.002019	0.991917	0.135597	0.251965
TDA-EN12	0.002019	0.991917	0.135600	0.251964
Vol-EN11	0.008967	0.823876	0.038379	0.216935
Beta-EN11	0.013042	0.913422	0.323293	0.408621
BetaVol	0.010426	0.933884	0.283281	0.372776
BetaVol-Long	0.010426	0.933884	0.283281	0.372776

Table 4: Out-of-sample performance: Period II — Risk and downside measures

Model	MaxDD	DownsideDev	VaR	CVaR
TE	-0.361383	0.002487	0.022835	0.039050
TDA-Lasso	-0.353294	0.001334	0.022999	0.038670
TDA-EN11	-0.353310	0.001334	0.023000	0.038670
TDA-EN12	-0.353294	0.001334	0.022999	0.038670
Vol-EN11	-0.383302	0.006420	0.016097	0.033865
Beta-EN11	-0.434419	0.008342	0.037309	0.057913
BetaVol	-0.393007	0.006602	0.034049	0.052779
BetaVol-Long	-0.393007	0.006601	0.034048	0.052779

The results show that the TDA-based models remain the best models for strict index tracking. In both periods, TDA-Lasso, TDA-EN11, and TDA-EN12 produce the lowest tracking errors and the highest correlations with the index. In Period I, TDA-EN11 has the lowest tracking error of 0.001626, while in Period II all three TDA models produce a tracking error of about 0.002019 and a correlation of 0.991917.

The economic models behave differently. BetaVol and BetaVol-Long do not beat the TDA models in tracking accuracy, but they produce stronger annualised returns. In Period I, BetaVol has an annualised return of 0.186003 compared with 0.126535 for TDA-EN11. In Period II, BetaVol has an annualised return of 0.283281 compared with 0.135597 for TDA-EN11. However, this higher return comes with higher volatility, higher tracking error, and larger downside risk. In Period I, BetaVol has a VaR of 0.027882 and CVaR of 0.044375, compared with 0.016650 and 0.027583 for TDA-EN11. Similarly, in Period II, BetaVol has a VaR of 0.034049 and CVaR of 0.052779, compared with 0.023000 and 0.038670 for TDA-EN11. This shows that BetaVol produces stronger returns, but exposes the portfolio to larger tail losses. Therefore, BetaVol should not be interpreted as a better strict tracker. Rather, it is better described as an interpretable enhanced sparse-tracking model that accepts greater downside risk in exchange for higher return potential.

The VaR and CVaR results also show that lower volatility does not automatically imply better tail-risk control. For example, Vol-EN11 has the lowest volatility in both periods and also produces the lowest VaR and CVaR values. However, its tracking error is much higher and its correlation with the index is much lower than the TDA models. This means that Vol-EN11 reduces portfolio risk partly by moving away from the benchmark rather than by tracking the index more efficiently. By contrast, the TDA models maintain low tracking error, high benchmark correlation, and relatively controlled VaR and CVaR. This supports the view that the TDA penalties are more suitable for strict benchmark replication, while the economic models introduce a clearer trade-off between return, tracking accuracy, and downside risk.

The similarity between BetaVol and BetaVol-Long is also important. Their results are almost identical in both periods, which suggests that the unconstrained BetaVol model is not relying heavily on short-selling. This supports the practical relevance of the BetaVol specification.

5.2 Growth of One Dollar

Figures 1 and 2 present the growth of \$1 for Period I and Period II. These plots compare the wealth paths of the different models relative to the S&P 500 benchmark.

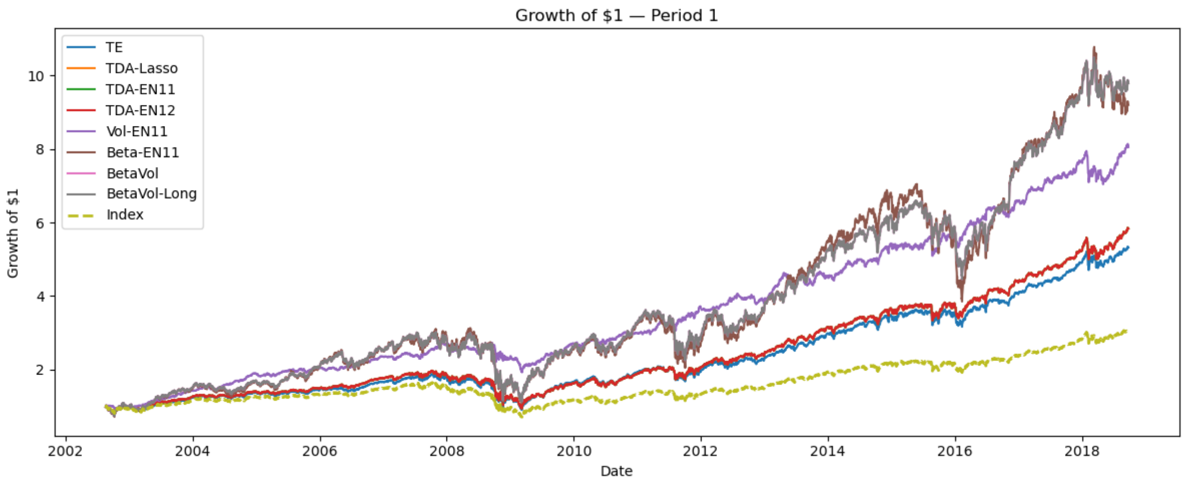

Figure 1: Growth of \$1: Period I

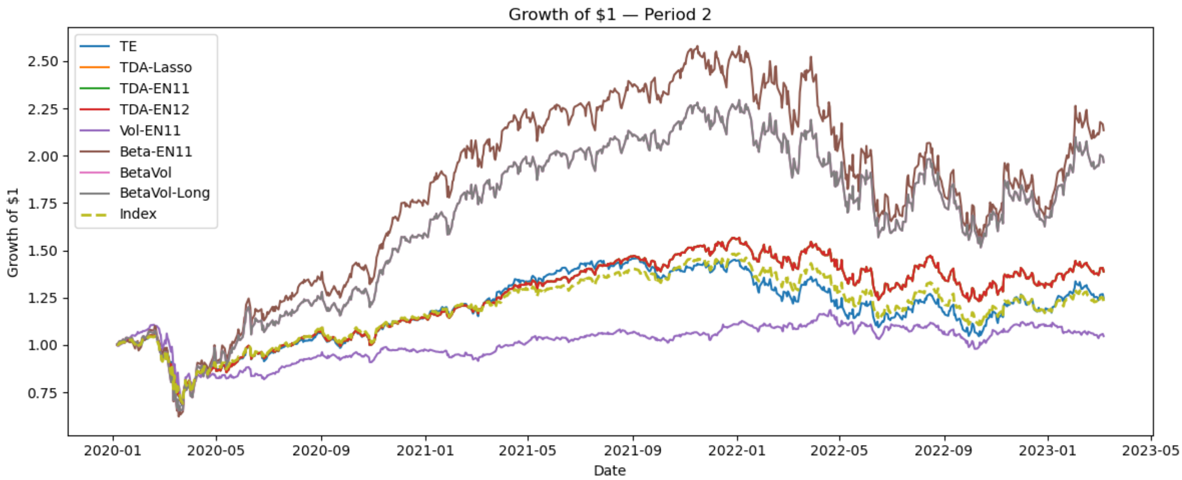

Figure 2: Growth of \$1: Period II

The growth plots should be interpreted together with the tracking error and risk measures. A model may generate a higher final wealth value but still be a weaker index tracker if it has a higher tracking error and lower correlation with the benchmark. This pattern is observed for the BetaVol model. Although BetaVol produces stronger return performance, it does not follow the index as closely as the TDA-based models. Therefore, its performance should be interpreted as return-enhanced tracking rather than pure benchmark replication.

5.3 Penalty Correlation Analysis

Tables 5 and 6 compare the TDA landscape norms with the economic penalty proxies. This helps show whether the TDA penalties and the economic penalties are capturing similar information.

Table 5: Correlation between TDA and economic penalties: Period I

Comparison	**Pearson Corr**	**Spearman Corr**	**N**
TDA H_0 alpha vs economic inverse-beta alpha	-0.162177	-0.535917	68515
TDA H_1 beta vs economic volatility penalty	0.587414	0.573882	68515

Table 6: Correlation between TDA and economic penalties: Period II

Comparison	**Pearson Corr**	**Spearman Corr**	**N**
TDA H_0 alpha vs economic inverse-beta alpha	-0.266915	-0.459126	18126
TDA H_1 beta vs economic volatility penalty	0.584570	0.533325	18126

The correlation results show that the TDA H_0 penalty and the inverse-beta penalty are negatively related. This means that the TDA sparsity penalty and the economic inverse-beta penalty do not rank stocks in the same way. In Period I, the Spearman correlation is -0.535917, while in Period II it is -0.459126. This suggests that the topological H_0 penalty captures information that is different from simple index beta.

By contrast, the TDA H_1 penalty has a moderate positive relationship with volatility. The Pearson correlation is 0.587414 in Period I and 0.584570 in Period II. This suggests that the H_1 landscape norm is more closely related to conventional volatility risk. However, the relationship is not perfect, so the TDA H_1 penalty still contains additional information beyond volatility.

5.4 Window-by-Window Penalty Comparison

To check whether the penalty relationship is stable over time, the correlations are also computed window-by-window. Tables 7 and 8 summarise the window-level correlations.

Table 7: Window-by-window penalty correlation summary: Period I

Statistic	Pearson: H0 vs Inv-Beta	Spearman: H0 vs Inv-Beta	Pearson: H1 vs Vol	Spearman: H1 vs Vol
Mean	-0.432232	-0.613456	0.682774	0.684220
Std	0.134387	0.085645	0.052724	0.075918
Min	-0.648069	-0.765029	0.446265	0.482223
25%	-0.568689	-0.682874	0.656541	0.632140
50%	-0.411681	-0.615279	0.698376	0.693461
75%	-0.372141	-0.544333	0.719261	0.739981
Max	0.010205	-0.401559	0.763095	0.814272

Table 8: Window-by-window penalty correlation summary: Period II

Statistic	Pearson: H0 vs Inv-Beta	Spearman: H0 vs Inv-Beta	Pearson: H1 vs Vol	Spearman: H1 vs Vol
Mean	-0.299453	-0.507887	0.638509	0.623302
Std	0.086700	0.118693	0.089575	0.080699
Min	-0.458138	-0.675322	0.396913	0.372387
25%	-0.337478	-0.609830	0.594075	0.588931
50%	-0.299926	-0.480098	0.635814	0.618817
75%	-0.254824	-0.452259	0.701618	0.674201
Max	-0.083333	-0.173147	0.772476	0.734500

The window-level results confirm the pooled correlation findings. The relationship between TDA H_0 and inverse beta is generally negative across time, while the relationship between TDA H_1 and volatility is consistently positive. This supports the idea that TDA and economic penalties do not measure the exact same type of risk.

5.5 Epsilon Sensitivity

The BetaVol model depends on the stabilisation constant ε in the inverse-beta mapping. Tables 9 and 10 report the sensitivity of BetaVol across four values of ε.

Table 9: Epsilon sensitivity analysis: Period I

ε	**TError**	**Correlation**	**Ann.Return**	**Volatility**	**MaxDD**	**VaR**
10^{-8}	0.008579	0.946875	0.186003	0.296954	-0.623272	0.027882
10^{-6}	0.008579	0.946875	0.186003	0.296954	-0.623272	0.027882
10^{-4}	0.008579	0.946875	0.186004	0.296957	-0.623271	0.027881
10^{-2}	0.008585	0.946822	0.185933	0.297033	-0.623242	0.027876

Table 10: Epsilon sensitivity analysis: Period II

ε	**TError**	**Correlation**	**Ann.Return**	**Volatility**	**MaxDD**	**VaR**
10^{-8}	0.010426	0.933884	0.283281	0.372776	-0.393007	0.034049
10^{-6}	0.010426	0.933884	0.283281	0.372776	-0.393007	0.034049
10^{-4}	0.010426	0.933884	0.283278	0.372776	-0.393009	0.034049
10^{-2}	0.010429	0.933894	0.283005	0.372847	-0.393200	0.034032

The sensitivity results show that the BetaVol model is stable across the tested values of ε. The tracking error, correlation, annualised return, volatility, drawdown, and VaR change only slightly as ε moves from 10^{-8} to 10^{-2}. This suggests that the results are not driven by one arbitrary choice of the stabilisation constant.

5.6 VaR Backtesting: Kupiec Test

The Kupiec test evaluates whether realised VaR violations are statistically consistent with the selected confidence level.

Table 11: Panel I Kupiec VaR backtesting results

Model	Kupiec p-value	Decision
TE	0.7547	Fail to Reject
TDA-Lasso	0.5499	Fail to Reject
TDA-EN11	0.5499	Fail to Reject
Vol-EN11	0.0000	Reject
Beta-EN11	0.0964	Fail to Reject
BetaVol	0.1724	Fail to Reject
BetaVol-Long	0.1724	Fail to Reject

Source: Authors' construct (2026).

Table 12: Panel II Kupiec VaR backtesting results

Model	Kupiec p-value	Decision
TE	0.0599	Fail to Reject
TDA-Lasso	0.2017	Fail to Reject
TDA-EN11	0.2017	Fail to Reject
Vol-EN11	0.2017	Fail to Reject
Beta-EN11	0.3329	Fail to Reject
BetaVol	0.3329	Fail to Reject
BetaVol-Long	0.3329	Fail to Reject

Source: Authors' construct (2026).

The Kupiec backtesting results indicate that most models produce statistically acceptable VaR estimates across both periods. The proposed BetaVol framework consistently passes the Kupiec test in both panels, suggesting that the model generates reasonably calibrated downside risk estimates despite the increased market turbulence during the COVID-period.

In contrast, the Vol-EN11 model fails the Kupiec test in Panel I, indicating that its VaR estimates underestimate realised tail risk despite producing lower overall volatility. This highlights an important distinction between reducing variance and accurately capturing extreme downside events. Overall, the results suggest that the BetaVol framework achieves a relatively balanced combination of risk reduction and reliable downside risk estimation across both normal and turbulent market environments.

5.7 Risk-Adjusted Performance and Turnover

Risk-adjusted performance is evaluated using the Sharpe Ratio and Information Ratio, while implementation efficiency is assessed using turnover and the number of selected assets.

Table 13: Panel I risk-adjusted performance and turnover

Model	Sharpe Ratio	Information Ratio	Turnover	Number of Assets
TE	0.641256	1.328632	0.839791	355.00
TDA-Lasso	0.689648	1.547456	0.395900	117.70
TDA-EN11	0.689641	1.547657	0.395762	117.73
Vol-EN11	1.112665	0.426070	0.190349	18.19
Beta-EN11	0.585553	0.607703	0.208271	33.99
BetaVol	0.626369	0.729372	0.144907	74.22
BetaVol-Long	0.626370	0.729375	0.144908	74.40

Source: Authors' construct (2026).

Table 14: Panel II risk-adjusted performance and turnover

Model	Sharpe Ratio	Information Ratio	Turnover	Number of Assets
TE	0.397612	0.102957	5.265359	477.00
TDA-Lasso	0.538170	1.191067	0.394761	127.08
TDA-EN11	0.538158	1.191006	0.394607	127.13
Vol-EN11	0.176914	-0.415289	0.206529	21.89
Beta-EN11	0.791181	1.092438	0.216767	43.79
BetaVol	0.759923	1.124527	0.159135	91.97
BetaVol-Long	0.759924	1.124529	0.159135	92.11

Source: Authors' construct (2026).

The results show clear differences between the models in terms of performance and implementation efficiency. In Panel I, the TDA models produce the highest Information Ratios, indicating that they track the benchmark more effectively after adjusting for active risk. However, they require more assets and have higher turnover than the BetaVol model. The Vol-EN11 model produces the highest Sharpe Ratio in the long-run period, suggesting strong risk reduction, but its lower Information Ratio shows that this comes at the expense of weaker benchmark tracking.

The BetaVol model has substantially lower turnover than the TDA and TE models, which implies lower transaction costs and easier implementation. It also uses fewer assets than the TDA models while maintaining reasonable tracking performance. In Panel II, which covers the COVID and recent period, BetaVol achieves a higher Sharpe Ratio than the TDA models while keeping turnover low. This suggests that the Beta–Volatility penalty structure provides greater stability during uncertain market conditions.

Overall, the TDA models remain preferable when the main objective is tracking accuracy, while BetaVol offers a stronger balance between risk-adjusted performance, sparsity, turnover control, and implementation cost efficiency.

5.8 Summary of Findings

The results show a clear trade-off. The TDA models are better for strict index tracking because they achieve lower tracking error and higher correlation with the benchmark. This means that the persistence landscape penalties are useful for close benchmark replication.

The BetaVol model adds value in a different way. It provides a simple and interpretable economic penalty structure. Inverse beta is used to guide asset selection, while volatility is used to control weight size. The model produces higher annualised returns than the TDA models in both periods, but this comes with higher tracking error and higher volatility. Therefore, BetaVol is best interpreted as an enhanced sparse-tracking model, not as a direct replacement for TDA.

The penalty correlation analysis further shows that the TDA and economic penalties do not fully overlap. TDA H_0 behaves differently from inverse beta, while TDA H_1 is moderately related to volatility. This supports the view that TDA captures structural information that is not fully explained by standard financial risk measures.

6 Discussion

The empirical results show that the TDA-based models remain the strongest specifications when the main objective is strict index replication. Across both sample periods, the TDA-Lasso, TDA-EN11 and TDA-EN12 models produce the lowest tracking errors and the highest correlations with the S&P 500 index. This suggests that the persistence landscape penalties used in the original framework are effective at selecting assets that preserve the behaviour of the benchmark. In this sense, the replication supports the main conclusion of Goel et al. (2025): topological information can be useful in sparse index tracking because it allows the regularisation process to be guided by more than only standard return moments.

The comparison between the TDA models and the economic extensions also highlights an important trade-off. The BetaVol model does not improve tracking accuracy relative to the TDA models. Its tracking error is higher and its correlation with the index is lower in both periods. Therefore, it cannot be interpreted as a superior pure index tracking model. However, the BetaVol model generates higher annualised returns than the TDA specifications. This shows that the economic penalty structure changes the nature of the portfolio. Instead of producing the closest possible benchmark replication, it behaves more like an enhanced sparse-tracking model, where the portfolio still follows the index reasonably well but takes on more active exposure in search of higher returns.

This difference is especially clear when looking at the risk measures. The BetaVol and Beta-EN11 models produce higher volatility, larger drawdowns and higher VaR values than the TDA models. This means that the higher returns from the economic models are not free; they are obtained by accepting more risk and weaker benchmark alignment. The Vol-EN11 model, on the other hand, has lower volatility and VaR in some cases, but its tracking performance is much weaker. This suggests that penalising volatility alone may control risk, but it does not necessarily preserve the co-movement needed for effective index tracking. The combined BetaVol specification is therefore more balanced than using beta or volatility alone, but it still does not match the tracking accuracy of the TDA framework.

The penalty correlation results provide a useful explanation for why the models behave differently. The TDA H0 penalty is negatively related to the inverse-beta penalty, which means that the two approaches rank assets differently. In practical terms, the assets that appear important from a topological point of view are not necessarily the same assets that appear attractive based on index beta. This supports the view that the H0 persistence landscape captures structural information that is not contained in a simple linear beta measure. Since beta only measures linear sensitivity to the index, it may miss more complex features in the return series, such as clustering behaviour, local instability, or nonlinear patterns.

The relationship between the TDA H1 penalty and volatility is more positive and moderate. This is intuitive because both measures are connected to risk and instability in the asset return process. However, the correlation is not perfect, which means that H1 is not simply a complicated version of volatility. It still appears to contain additional information. This is important because it suggests that persistent homology may capture forms of return structure that are related to risk but not fully explained by traditional second-moment measures.

The window-by-window correlation analysis strengthens this interpretation. The negative relationship between H0 and inverse beta, and the positive relationship between H1 and volatility, are not isolated results from one period. They are broadly stable across rolling windows. This indicates that the difference between TDA-based and economic penalties is persistent over time. Therefore, the results are not only caused by one unusual market episode, but reflect a more general difference in what the two penalty systems measure.

Another important finding is that the BetaVol model is stable across different values of the stabilisation constant ε. The sensitivity analysis shows that changing ε from 10^{-8} to 10^{-2} has almost no effect on tracking error, correlation, return, volatility, drawdown or VaR. This is useful because it means the BetaVol results are not driven by an arbitrary numerical choice. The inverse-beta penalty is therefore stable enough to be used as an interpretable benchmark against the TDA model.

The similarity between BetaVol and BetaVol-Long is also meaningful. Since the long-only version produces almost identical results to the unconstrained BetaVol model, the economic extension does not appear to rely heavily on short positions. This improves the practical relevance of the model, because long-only portfolios are easier to implement and more realistic for many institutional investors. It also suggests that the beta-volatility penalty structure naturally produces portfolios that are close to long-only, even without explicitly imposing the constraint.

Overall, the results suggest that the TDA approach is better suited for the original goal of sparse index tracking. It gives closer replication, stronger benchmark correlation and more controlled tracking behaviour. The economic extension is useful, but for a slightly different purpose. Its main advantage is interpretability: beta has a clear meaning as index sensitivity, while volatility has a clear meaning as standalone asset risk. This makes the BetaVol model easier to explain and cheaper to compute than the TDA pipeline. However, this simplicity comes at the cost of losing some structural information that the TDA penalties appear to capture.

Therefore, the main conclusion from the replication and extension is not that one approach completely dominates the other. Rather, the choice depends on the objective. If the investor wants the closest sparse replication of the index, the TDA-based Elastic-Net models are preferred. If the investor wants a simpler, more interpretable model that still tracks the index reasonably well while allowing more active return exposure, the BetaVol model may be useful. The extension therefore adds value by showing that standard financial risk measures can provide a competitive and transparent benchmark, but they do not fully replace the topological information used in the original model.

7 Conclusion

This study replicated the TDA-based sparse index tracking framework of Goel et al. (2025) and extended it by replacing the topological penalties with CAPM beta and volatility penalties. The results show that the TDA-based models provide the best strict index tracking performance, with lower tracking errors and higher correlations with the S&P 500 across both periods. This supports the usefulness of persistence landscape penalties in sparse index replication.

The economic BetaVol extension provides a simpler and more interpretable alternative. Although it does not track the index as closely as the TDA models, it produces higher annualised returns and remains stable across different values of ε. Therefore, BetaVol is better viewed as an enhanced sparse-tracking model rather than a direct replacement for TDA.

Overall, the findings suggest that TDA captures structural information that is not fully explained by beta and volatility. However, standard financial risk measures still offer a useful benchmark because they are easier to interpret and cheaper to compute. Future work could combine TDA penalties with beta and volatility penalties in one hybrid model to test whether both structural and economic risk information can improve sparse index tracking performance.

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